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Java SE 27 & JDK 27

Class PhantomReference<T>

java.lang.Object
  java.lang.ref.Reference<T>
    java.lang.ref.PhantomReference<T>

Type Parameters:

T - the type of the referent

```
public non-sealed class PhantomReference<T>
extends Reference<T>
```

Phantom reference objects, which are enqueued after the collector determines that their referents may otherwise be reclaimed. Phantom references are most often used to schedule post-mortem cleanup actions.

Suppose the garbage collector determines at a certain point in time that an object is phantom reachable. At that time it will atomically clear all phantom references to that object and all phantom references to any other phantom-reachable objects from which that object is reachable. At the same time or at some later time it will enqueue those newly-cleared phantom references that are registered with reference queues.

In order to ensure that a reclaimable object remains so, the referent of a phantom reference may not be retrieved: The `get` method of a phantom reference always returns `null`. The `refersTo` method can be used to test whether some object is the referent of a phantom reference.

Since:

1.2

Constructor Summary

Constructors

Constructor	Description
`PhantomReference(T referent, ReferenceQueue<? super T> q)`	Creates a new phantom reference that refers to the given object and is registered with the given queue.

Method Summary

All Methods | **Instance Methods**

Modifier and Type	Method	Description
T	get()	

	Returns this reference object's referent.

Methods declared in class Reference

`clear`, `clone`, `enqueue`, `isEnqueued`, `reachabilityFence`, `refersTo`

Methods declared in class Object

`equals`, `finalize`, `getClass`, `hashCode`, `notify`, `notifyAll`, `toString`, `wait`, `wait`, `wait`

Constructor Details

PhantomReference

```
public PhantomReference(T referent,
                        ReferenceQueue<? super T> q)
```

Creates a new phantom reference that refers to the given object and is registered with the given queue.

It is possible to create a phantom reference with a `null` queue. Such a reference will never be enqueued.

Parameters:

`referent` - the object the new phantom reference will refer to

`q` - the queue with which the reference is to be registered, or `null` if registration is not required

Method Details

get

```
public T get()
```

Returns this reference object's referent. Because the referent of a phantom reference is always inaccessible, this method always returns `null`.

Overrides:

`get` in class `Reference<T>`

Returns:

`null`

See Also:

`Reference.refersTo(T)`

Report a bug or suggest an enhancement
For further API reference and developer documentation see the Java SE Documentation, which contains more detailed, developer-targeted descriptions with conceptual overviews, definitions of terms, workarounds, and working code examples. Other versions.
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